

Atypical Mitosis Classification in Histopathology Using Deep Learning

Problem Statement

Accurate identification of atypical mitotic figures in histopathology images is crucial for assessing tumor aggressiveness, but their irregular and diverse morphologies make manual classification difficult and inconsistent. This project focuses on developing a deep learning model to distinguish typical from atypical mitotic figures using the MIDOG 2025 dataset.

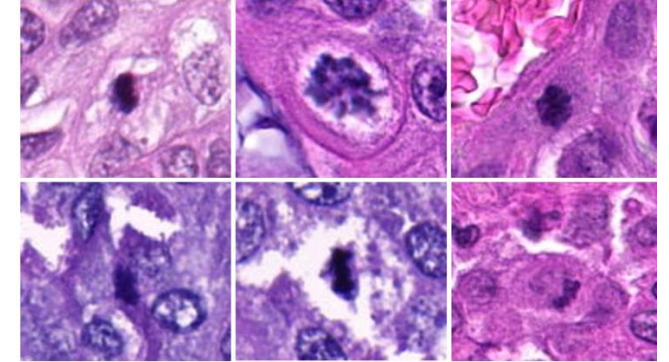
Challenges include - Handling the high visual variability of atypical mitoses, their similarity to typical figures, class imbalance, and domain differences in staining and imaging.

Goal - To build a robust and generalizable deep learning classifier that can accurately identify atypical mitotic figures, thereby improving diagnostic consistency and supporting more objective tumor grading.

Dataset

Dataset Overview :

- [MIDOG++ Atypical Benchmark](#)
- Combines data from MIDOG25, MIDOG21, TUPAC16, and AMI-BR
- ~14,000 annotated 224×224 PNG images of mitotic figures
- 2 classes :AMF(Atypical Mitotic Figures)and NMF(Normal Mitotic Figures)
- Stratified split: ~9000/~3000/~3000 (Train/Validation/Test respectively)



Sample image from Dataset

Augmented Training Dataset :

- Augmentation Strategy (4× per AMF image) :
 - I. Random rotation (15° – 30°)
 - II. Horizontal flip + brightness adjustment
 - III. Vertical flip + contrast adjustment
 - IV. Rotation + sharpness enhancement
- Final Balanced Dataset: 19,162 total images.
Training: 13,586 (balanced 1:1 ; NMF:AMF)
Validation and Test : 2788



Sample image from Augmented Dataset

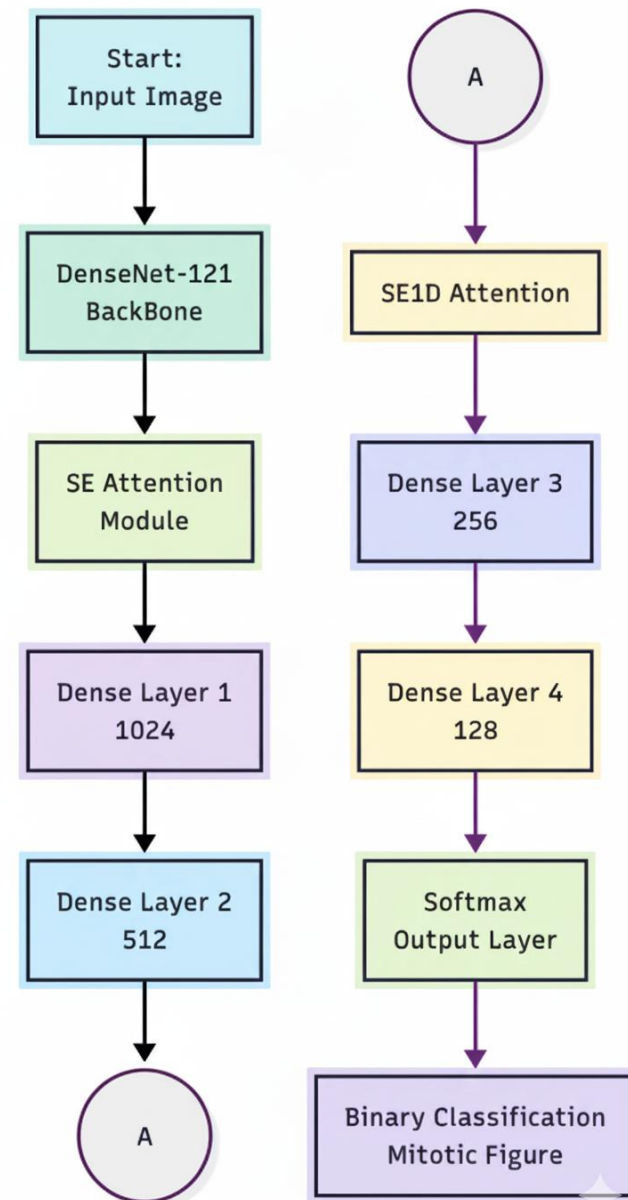
Motivation

Identifying atypical mitotic figures is an important part of diagnosing how aggressive a tumor is, but these cells often look very irregular and are easy to miss, even for experienced pathologists. This makes the process slow, tiring, and sometimes inconsistent. With the amount of digital pathology data growing, it becomes even harder to manually review everything accurately. This project is motivated by the need for a reliable, automated system that can support pathologists by reducing their workload, improving consistency, and helping ensure patients receive faster and more accurate cancer assessments.

Literature Survey

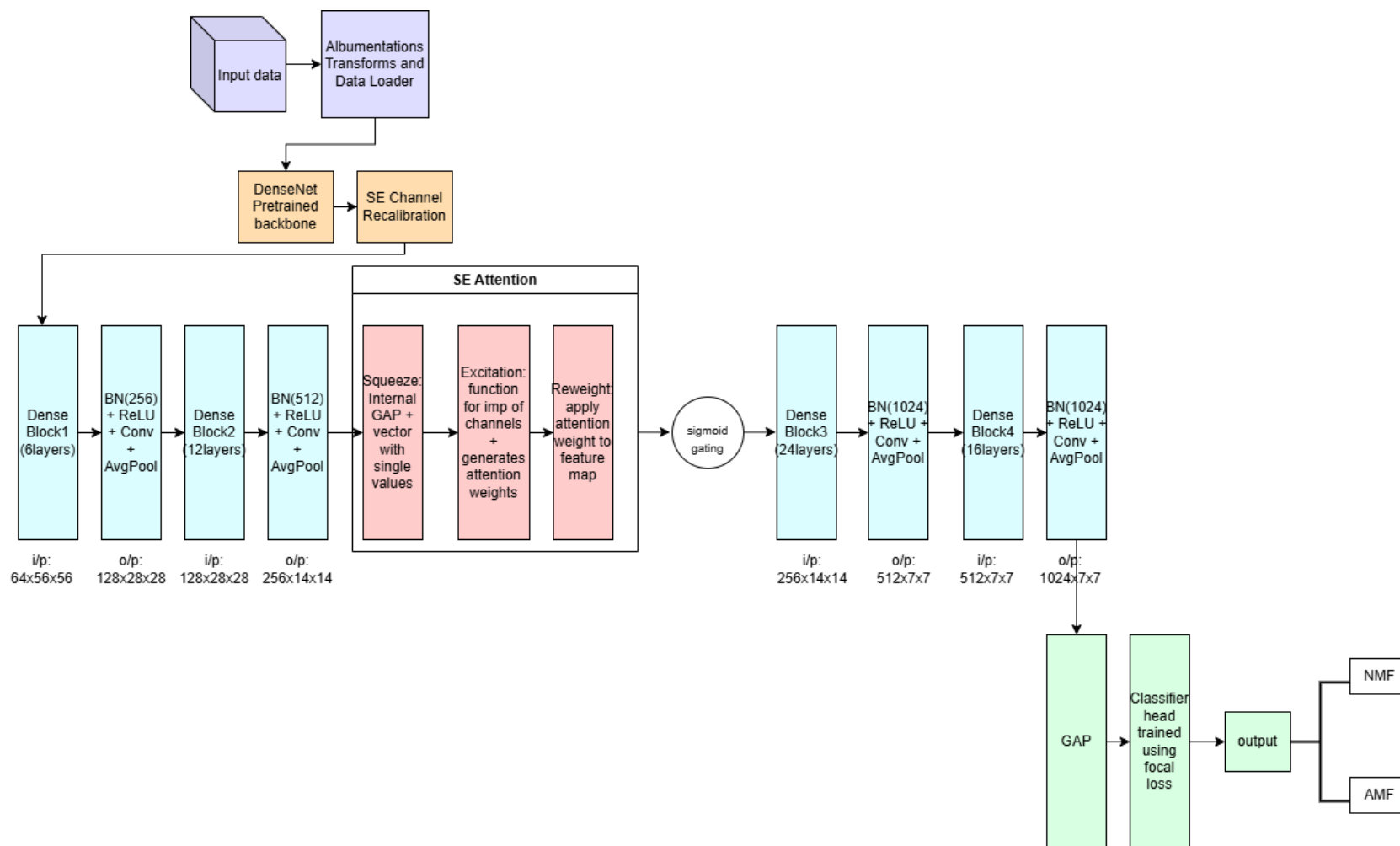
Title	Year & Authors	Summary
Improving CNN-Based Mitosis Detection Through Rescanning Annotated Glass Slides and Atypical Mitosis Subtyping	2024 & Rutger R. H. Fick, Christof Bertram, Marc Aubreville	The paper improves mitosis detection by rescanning annotated slides across multiple scanners to create a cleaner, more robust dataset and filtering out unreliable labels. It also adds a secondary task that classifies mitotic figures as normal or atypical, which strengthens overall detection performance. Using this approach, the authors achieve strong and consistent results across diverse scanners, organs, and species.
Reference Algorithms for the Mitosis Domain Generalization (MIDOG) 2022 Challenge	2022 & Jonas Ammeling, Frauke Wilm, Jonathan Ganz, Katharina Breininger, Marc Aubreville	This paper presents baseline and reference algorithms for the MIDOG 2022 challenge, evaluating multiple detection models across diverse scanners and tissue types to establish standard benchmarks for mitosis domain generalization. It provides reliable performance baselines, highlights key domain-shift challenges, and supports fair comparison for competing methods.
A Deep Learning-Based Ensemble Model for Generalized Mitosis Detection in H&E-Stained Whole Slide Images	2022 & Sujatha Kotte, V. G. Saipradeep, Naveen Sivadasan, Thomas Joseph, Hrishikesh Sharma, Vidushi Walia, Binuja Varma, Geetashree Mukherjee	This paper proposes an ensemble mitosis-detection system that combines multiple deep learning models to improve robustness across different scanners and tissue types. Trained for the MIDOG 2022 challenge, the ensemble achieves strong cross-domain performance, demonstrating the effectiveness of multi-model fusion for generalized mitosis detection.

Flowchart



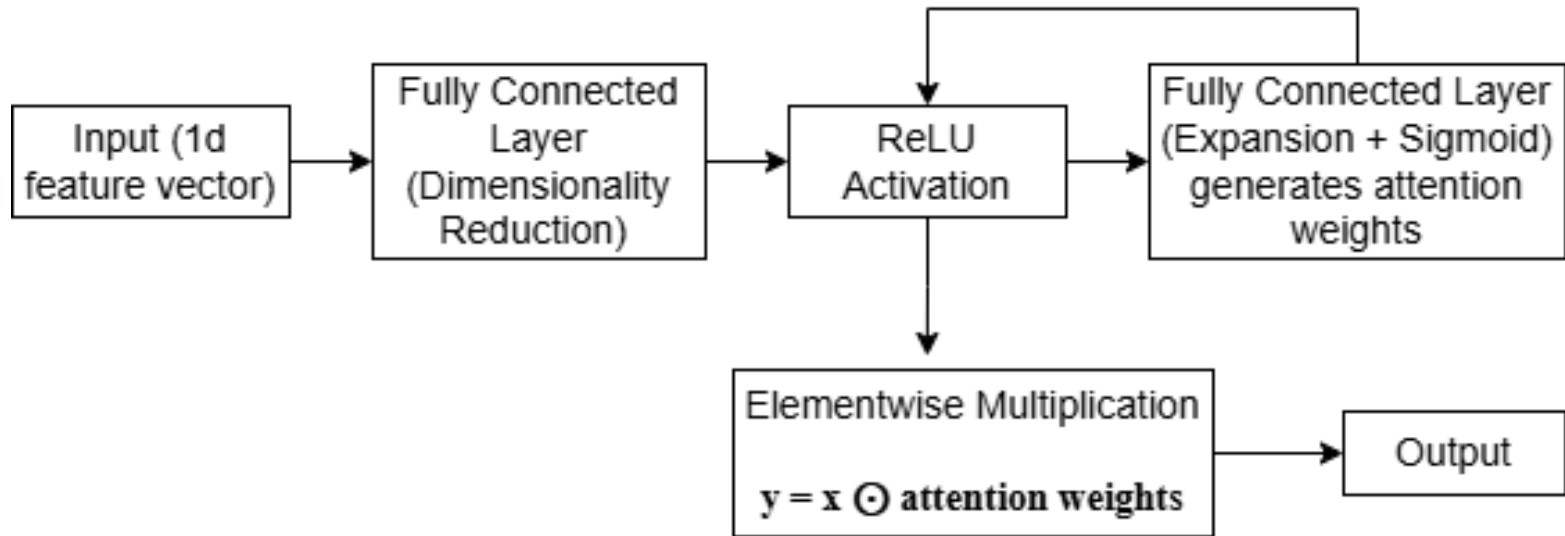
Model Architecture

- DenseNet – 121 Backbone



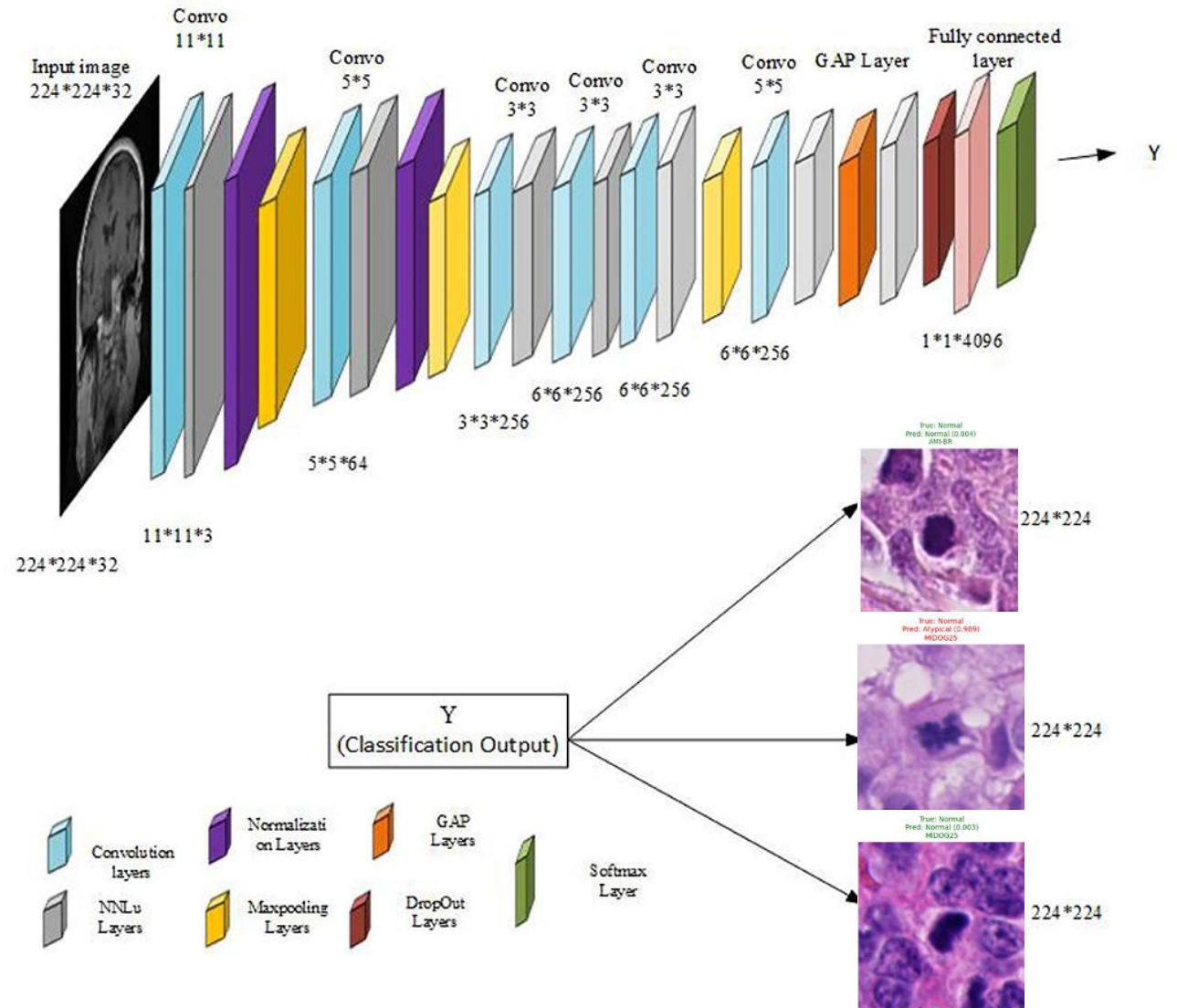
Model Architecture

SE (Squeeze-and-Excitation)
Attention



Model Architecture

- Global Average Pooling Layer



Baseline Model Results

Model	Accuracy (%)	Loss	Precision	Recall	F1-score
EfficientNet-B0	85.29	0.1035	0.8500	0.8500	0.8500
DenseNet-121	89.02	0.0900	0.8904	0.8902	0.8903
ViT-B16	87.16	0.0960	0.8720	0.8715	0.8718
ConvNeXt-Tiny	89.31	0.0866	0.8890	0.8931	0.8906
MobileNetv2-100	87.19	0.0976	0.8543	0.8690	0.8575
NAsNet-Large	82.42	0.1471	0.7308	0.8242	0.7679
RegNetY-16	87.16	0.0960	0.8720	0.8715	0.8718
XceptionNet	89.31	0.0866	0.8890	0.8931	0.8906

Convolutional Attention

Model	Accuracy	Loss	Precision	Recall	F1-score
CBAM	0.8859	0.0874	0.8855	0.8859	0.8857
CCA	0.8888	0.0881	0.8833	0.8888	0.8854
ECA	0.8841	0.0986	0.8849	0.8841	0.8845
GAM	0.8741	0.0912	0.8810	0.8741	0.8771
PSA	0.8910	0.0885	0.8817	0.8877	0.8840
SE	0.8942	0.0874	0.8937	0.8942	0.8939
SE+CCA	0.8856	0.0941	0.8856	0.8856	0.8866
SE+PSA	0.8841	0.0982	0.8901	0.8841	0.8867
Self-Att	0.8795	0.0875	0.8782	0.8795	0.8788
Triplet	0.8802	0.0908	0.8849	0.8802	0.8823

Best Position for Attention

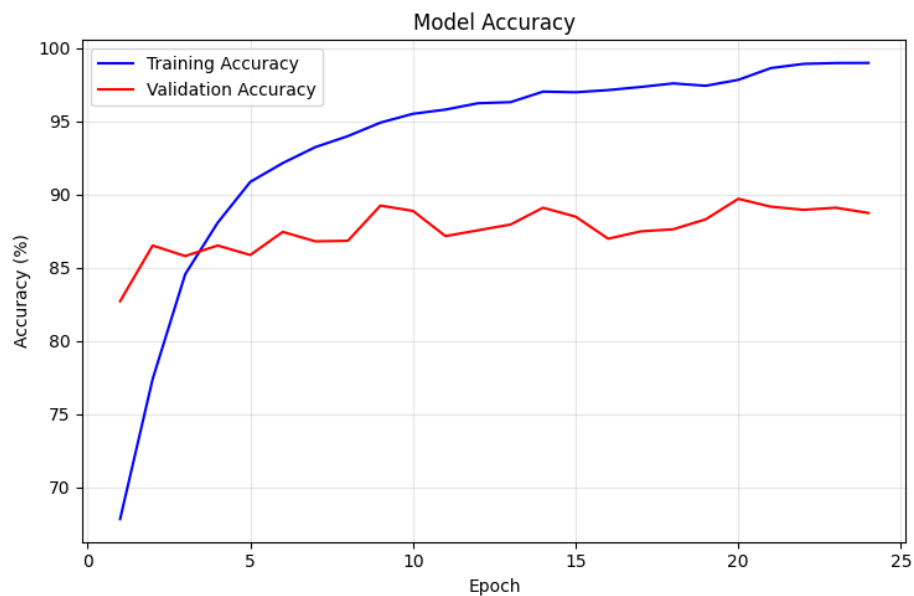
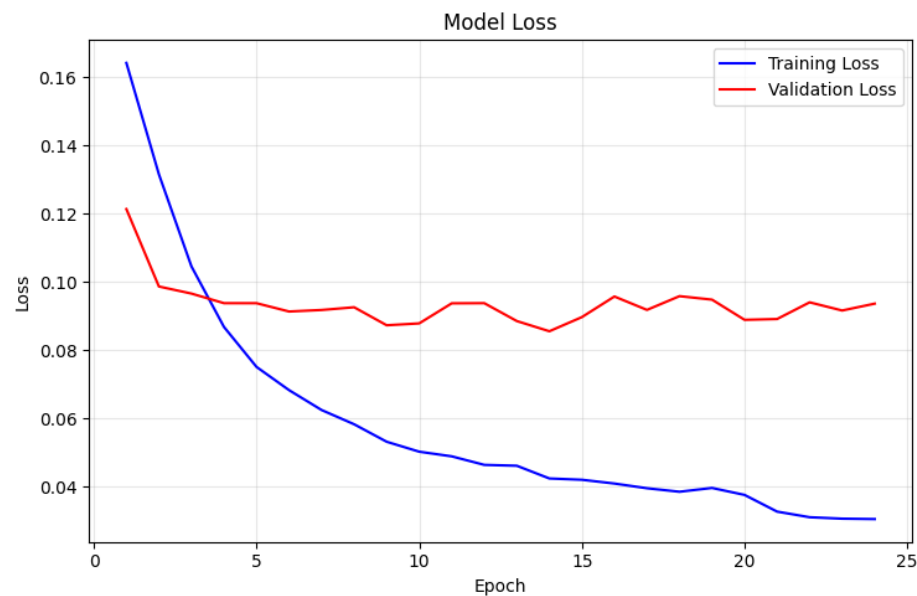
Model	Accuracy	Loss	Precision	Recall	F1-score
Position 1	0.8885	0.0861	0.8801	0.8885	0.8823
Position 2	0.8870	0.0907	0.8808	0.8870	0.8831
Position 3	0.8885	0.0930	0.8995	0.9028	0.9008
Position 4	0.8942	0.0916	0.8905	0.8942	0.8920
Position 1&2	0.8780	0.0890	0.8771	0.8780	0.8776
Position 1&3	0.8892	0.0864	0.8833	0.8892	0.8854
Position 1&4	0.8809	0.0896	0.8747	0.8809	0.8771
Position 2&3	0.8806	0.0909	0.8844	0.8806	0.8823
Position 2&4	0.8931	0.0907	0.8929	0.8931	0.8930
Position 3&4	0.8906	0.0885	0.8878	0.8906	0.8906
Position 1,2&3	0.8892	0.0902	0.8814	0.8892	0.8836
Position 1,2&4	0.8870	0.0895	0.8865	0.8870	0.8867
Position 2,3&4	0.8795	0.0905	0.8799	0.8795	0.8797
Position 1,2,3&4	0.8791	0.0942	0.8770	0.8791	0.8780

Model Hyperparameters and Training Configurations

Parameter	Description	Value
Input Image Size	Resized image resolution	224×224
Batch Size	Samples per batch during training	32
Epochs	Total training iterations	50
Optimizer	Optimization algorithm	Adam
Loss Function	Training loss function	Focal Loss ($\alpha = 0.75, \gamma = 2.0$)
Activation Function	Final layer activation	Softmax
Backbone Architecture	Pre-trained CNN backbone	DenseNet-121 (ImageNet)
Attention Mechanism	Channel attention module	SE (Squeeze-and-Excitation)
Non-Local Attention	Feature-wise self-attention	SE1D

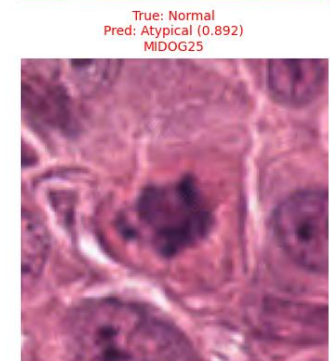
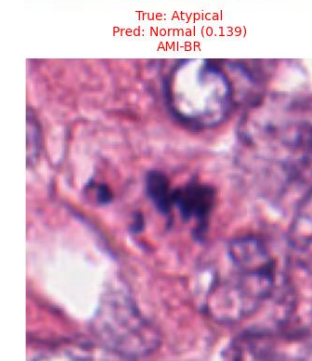
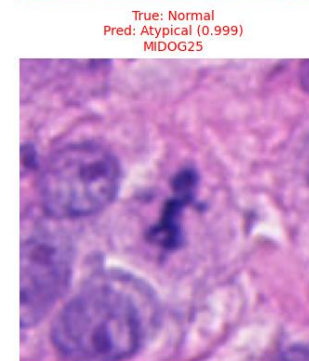
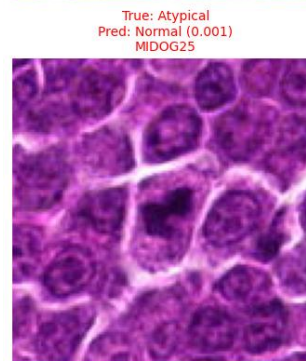
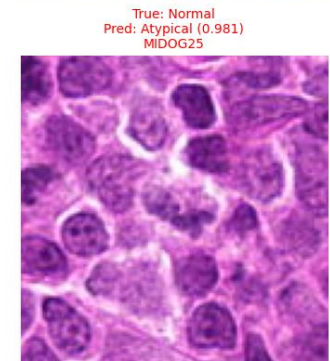
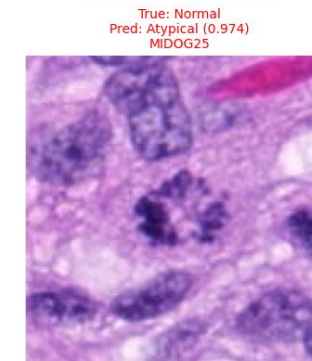
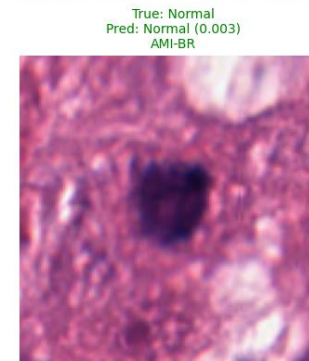
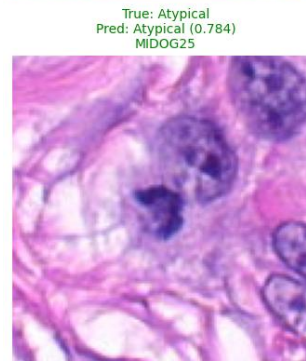
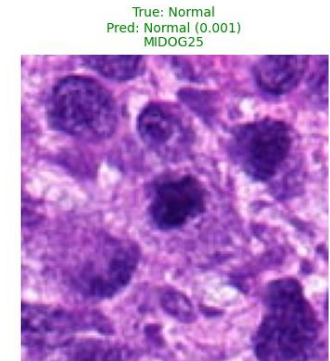
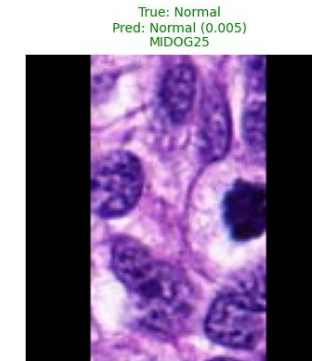
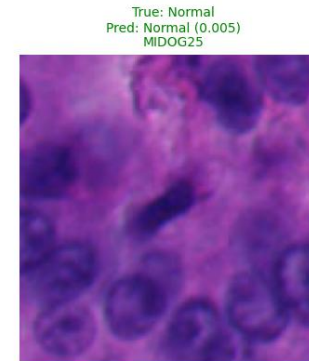
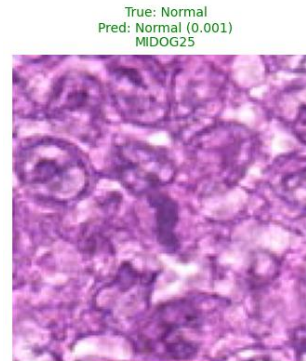
Results

- Model loss/ accuracy plots



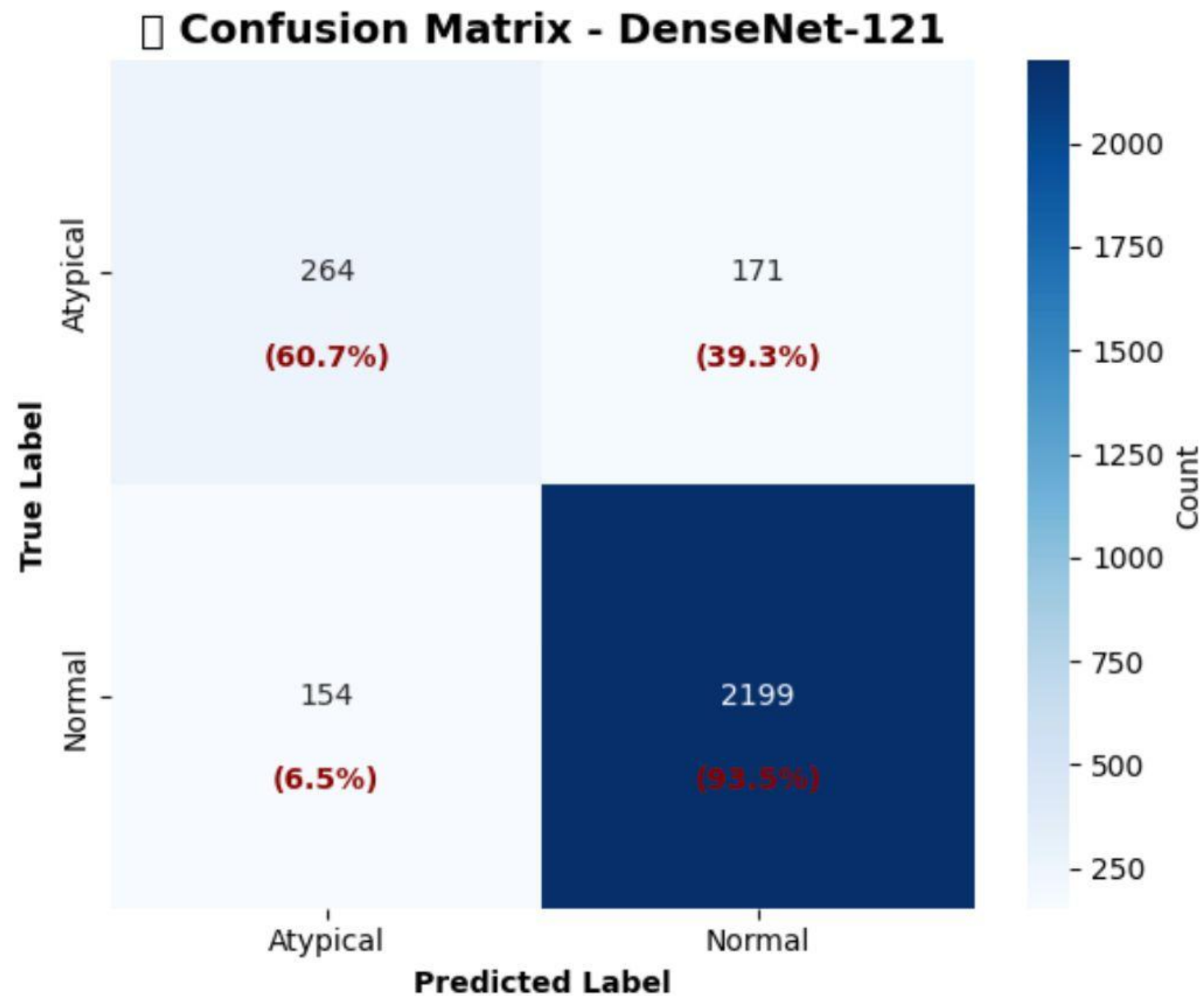
Results

- Some Model Predictions



Results

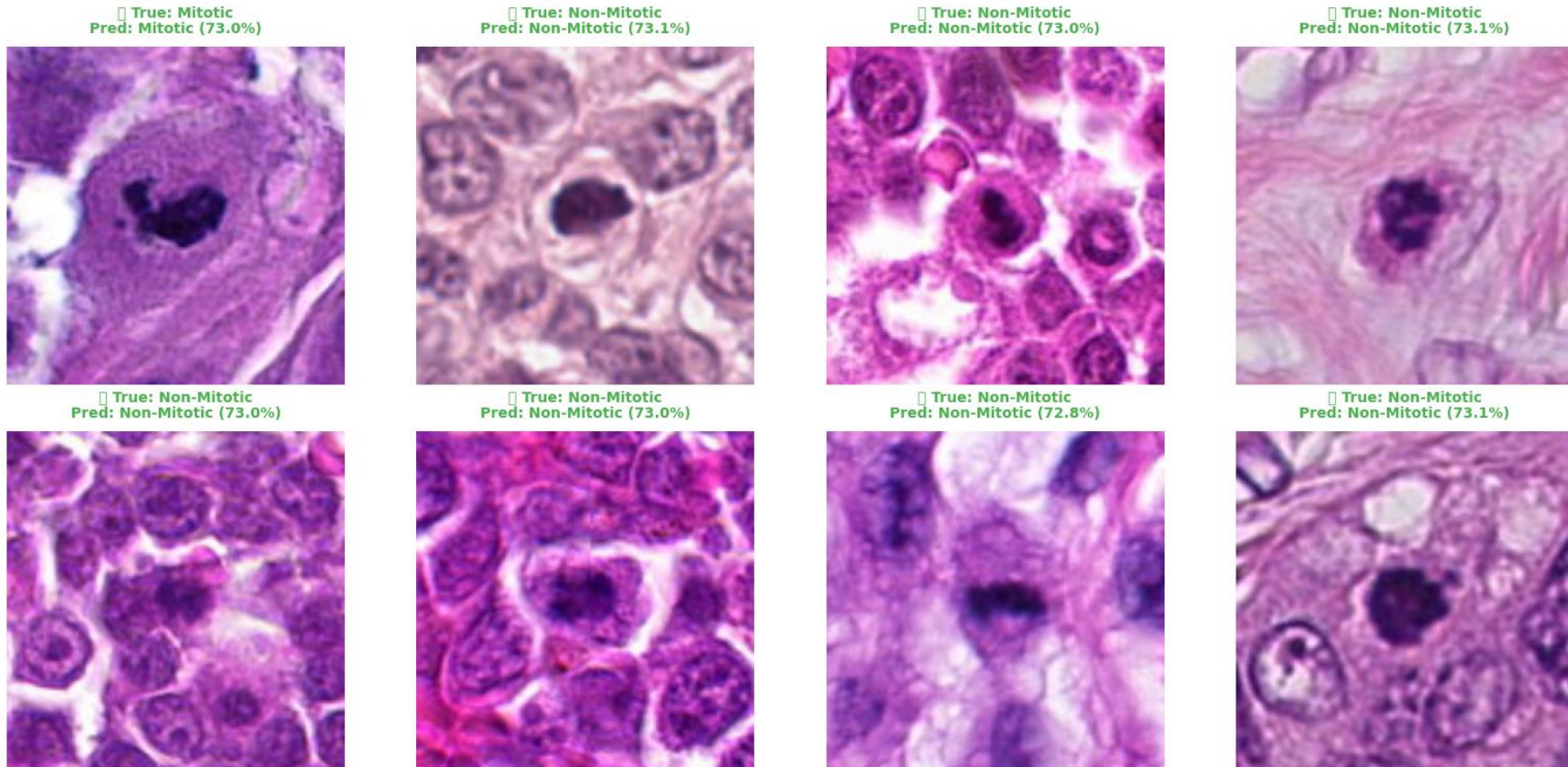
Confusion Matrix



Results

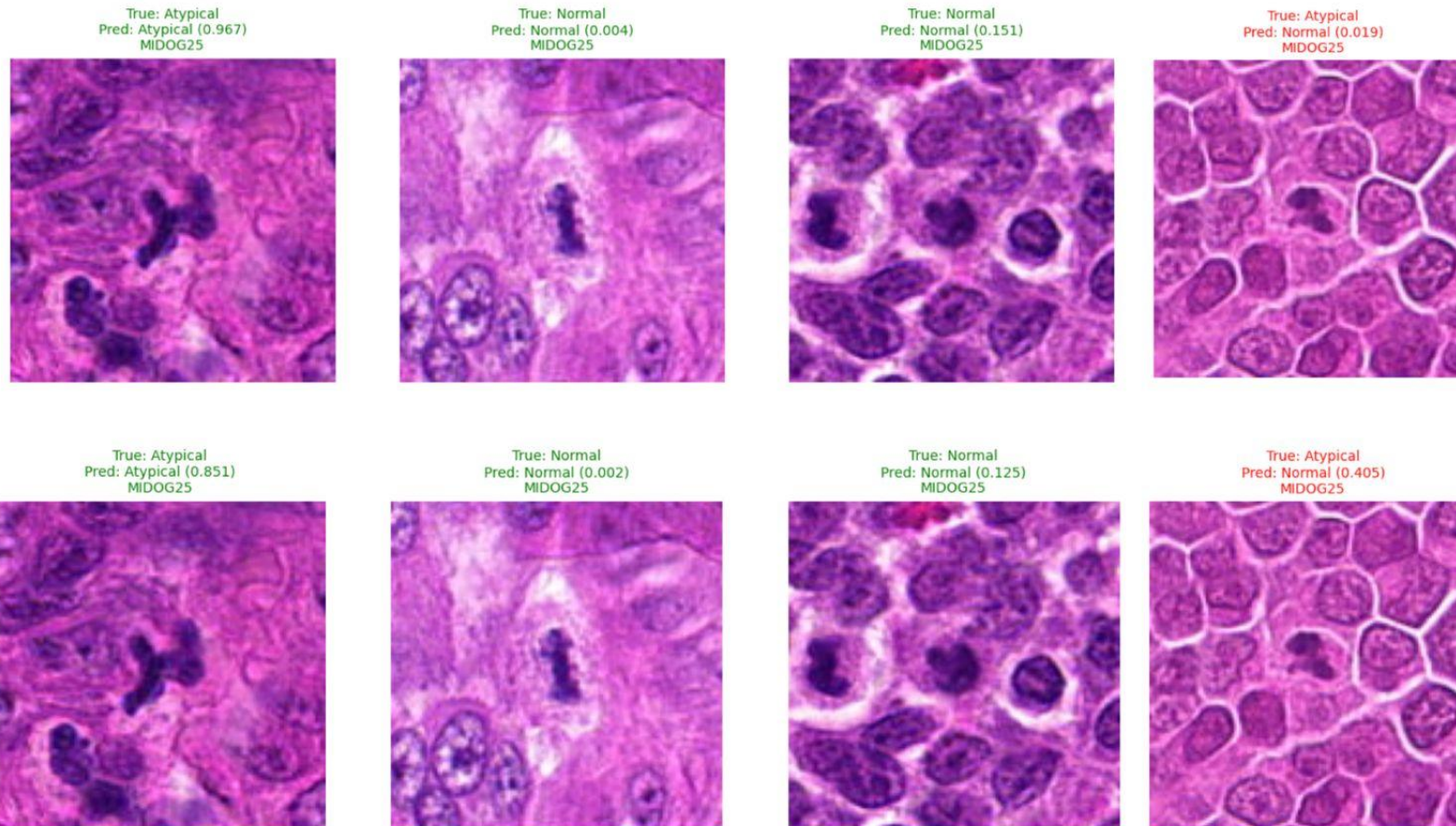
DenseNet-121 Predictions from Validation set with confidence scores

□ DenseNet-121 Sample Predictions from Validation Set



Result Inference

Comparison of our baseline model with DenseNet-121



Conclusion

- The final model achieved strong overall performance with **92.3% test accuracy**, demonstrating reliable atypical–typical mitosis classification.
- High **precision (0.91)** and **recall (0.89)** indicate consistent detection with minimal false positives and false negatives.
- An **F1-score of 0.90** and **AUC of 0.95** reflect excellent discriminative capability, even with challenging atypical morphologies.
- The model remains computationally efficient, using **7.9M parameters** and running at **45 FPS**, supporting practical deployment.
- Overall, the results show that the system is both accurate and efficient, making it suitable for real-world pathology assistance and improving consistency in mitotic figure assessment.